Srividya Bandari

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SUMMARY

Software Engineer with 3+ years of experience building distributed systems, cloud-native services, event-driven platforms, and production reliability and observability systems. Experienced in designing and operating high-scale C#/.NET and Java services across cloud infrastructure, streaming, observability, and AI-enabled applications. Delivered measurable improvements including 35% lower workflow latency and 70% lower p95 lookup latency.

WORK EXPERIENCE

Software Engineer **Jul 2023 – Present**
OPENLANE — *Tech: C#, .NET Core, ASP.NET Core, Azure, Apache Pulsar, Redis, PostgreSQL, Terraform* *Austin, TX*

- Owned and operated core C#/.NET Core services in a distributed inspection workflow platform processing 300K+ vehicle inspections and 3.5M+ validation, integration, audit, and observability events per day across inspection, inventory, and VIN intelligence systems
- Designed and shipped Apache Pulsar-based event-driven workflows with replay-safe consumers, correlation IDs, schema-versioned payloads, and failure-isolated retry paths, reducing workflow latency by 35% while improving real-time state consistency
- Improved distributed-system reliability by implementing idempotent consumers, deduplication keys, exponential backoff, dead-letter handling, and downstream failure isolation, preventing duplicate processing and invalid state transitions during service degradation
- Optimized high-volume VIN decoding and reference-data lookups using Redis caching, TTL-based invalidation, and request coalescing, reducing p95 latency by 70% and lowering repeated third-party service calls
- Led production rollout of .NET services on Azure using Terraform-managed infrastructure, staged enablement, canary deployments, rollback mechanisms, and Honeycomb distributed tracing, enabling faster issue detection and safer releases
- Built ML-assisted inspection decision workflows using VIN intelligence, vehicle metadata, inspection signals, and historical transition features to support anomaly detection, duplicate detection, risk scoring, and manual-review routing

Research Software Engineer Intern **May 2022 – Aug 2022**
Mayo Clinic / ASU — *Tech: Python, PyTorch, TensorFlow, Multi-view CNNs, SageMaker, Datadog* *Phoenix, AZ*

- Implemented a multi-view CNN combining X-ray images and patient metadata to predict osteoporosis risk, improving model accuracy from 87% to 95% and strengthening identification of high-risk cases
- Built preprocessing, enrichment, and validation pipelines for clinical imaging and patient data, reducing data-processing errors by 25% before model training and inference
- Modularized model training, evaluation, and inference components for integration with Mayo Clinic systems, enabling reuse across related clinical prediction workflows

Software Engineer Intern **Nov 2020 – Jun 2021**
Infosys — *Tech: Java, Python, Go, Spring Boot, Kafka, RabbitMQ, Redis, AWS, Docker, Jenkins, OpenSearch* *Hyderabad, India*

- Optimized the Lead Verifier CI across 150+ repositories by implementing SHA-based change detection and Redis-backed Docker image deduplication, reducing redundant builds by 35%, image duplication by 50%, and saving 1,000+ build hours per month
- Developed Jenkins pipelines and Python automation to synchronize codebases from Qualcomm repositories into client systems and automatically provision compatible repositories
- Integrated RabbitMQ, Logstash, and OpenSearch to collect, process, and visualize build metrics and logs across infrastructure supporting 170+ concurrent builds

PROJECTS

RAG-Powered Document Q&A System | *React, Python, FastAPI, FAISS, Docker, Vector Search*

- Built an end-to-end RAG application with a React frontend and FastAPI backend, supporting PDF upload, document indexing, semantic search, and source-grounded responses
- Implemented PDF ingestion, semantic chunking, FAISS retrieval, context ranking, and grounded Q&A; added retrieval quality checks that reduced irrelevant responses by 40% versus baseline generation

Multi-Threaded File System Simulator | *C++, POSIX-Style APIs, Read-Write Locks, Concurrency*

- Built an in-memory POSIX-style file-system simulator in C++ supporting inode-based metadata, block allocation, concurrent file operations, and read-write locking; improved throughput by 70% over a single-threaded baseline while processing 100,000 mixed read/write operations across eight threads

TECHNICAL SKILLS

Languages: C#, Java, Python, C++, JavaScript, TypeScript, SQL, Go
Backend: .NET Core, ASP.NET Core, Spring Boot, REST, gRPC, Microservices, Distributed Systems
Frontend: React
Cloud & Infrastructure: Azure, AWS, Docker, Kubernetes, Terraform, CI/CD, GitHub Actions, Jenkins
Observability: OpenTelemetry, Honeycomb, Datadog, Grafana, Prometheus, Splunk
Streaming & Messaging: Apache Pulsar, Kafka, RabbitMQ, Flink, Event-Driven Architecture
Databases & Caching: PostgreSQL, SQL Server, Redis, MongoDB, Elasticsearch, OpenSearch
AI & ML: Generative AI, LLMs, RAG, Vector Search, FAISS, PyTorch, TensorFlow, SageMaker

EDUCATION

Arizona State University – Master of Science, Computer Engineering Aug 2021 – May 2023
Osmania University – Bachelor of Engineering, Electrical and Electronics Engineering Aug 2016 – Jul 2020